

LEADER, SK, Canada

WMO: 714590

Lat: 50.910N

Lon: 109.503W

Elev: 2216

StdP: 13.56

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.6	-19.9	-33.5	0.9	-26.6	-26.2	1.4	-19.8	29.9	28.7	27.2	24.9	6.3	250	0.720

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.1	90.4	62.8	86.7	62.0	82.8	60.8	66.7	81.9	64.8	79.4	63.1	77.4	11.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.9	89.9	71.4	59.8	83.2	69.4	57.8	77.4	67.8	32.6	81.9	31.0	79.4	29.7	77.7	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.1	23.5	20.7	DB	-33.1	96.6	5.2	3.4	-36.8	99.0	-39.8	101.0	-42.7	102.9	-46.5	105.3
			WB	-33.3	70.6	5.1	2.2	-37.0	72.2	-40.0	73.5	-42.8	74.7	-46.6	76.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	39.5	10.7	13.7	26.5	41.0	52.3	60.4	66.3	65.0	55.4	41.2	25.8	13.5
	DBStd	22.87	16.20	14.58	14.88	9.42	8.20	6.24	5.57	6.32	8.34	9.51	13.20	15.22
	HDD50	5524	1219	1016	730	289	72	4	0	39	296	726	1133	
	HDD65	9543	1684	1436	1195	719	400	163	51	80	304	738	1175	1598
	CDD50	1675	0	0	1	20	143	317	505	464	200	24	1	0
	CDD65	217	0	0	0	0	6	26	91	79	15	0	0	0
	CDH74	3931	0	0	1	18	227	464	1397	1369	438	17	0	0
	CDH80	1423	0	0	0	1	51	126	525	561	158	1	0	0
	WSAvg	10.0	9.7	9.4	10.5	11.4	11.4	10.4	9.2	8.9	9.8	10.3	9.8	9.6
	PrecAvg	14.4	0.7	0.7	0.7	1.1	1.7	2.3	2.0	1.7	1.4	0.6	0.6	0.8
Precipitation	PrecMax	19.1	1.6	2.0	1.9	3.7	3.7	6.4	4.4	4.4	3.8	2.3	1.6	1.6
	PrecMin	10.6	0.1	0.1	0.0	0.1	0.2	0.6	0.5	0.2	0.0	0.0	0.1	0.1
	PrecStd	2.0	0.4	0.5	0.4	0.8	0.8	1.5	1.1	1.1	1.1	0.6	0.4	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.3	45.1	62.9	76.7	85.5	89.0	94.9	95.2	90.9	76.4	61.4	48.8
		MCWB	39.1	39.0	47.8	52.7	57.7	62.3	65.5	63.0	60.4	54.8	47.1	39.7
	2%	DB	40.1	39.7	55.6	69.8	80.1	83.4	90.0	90.7	84.6	68.4	53.2	42.3
		MCWB	35.5	35.2	43.6	49.9	55.8	60.5	64.9	61.7	58.6	51.5	43.0	36.6
	5%	DB	37.0	36.8	49.9	64.3	75.4	79.3	85.9	86.7	78.8	62.5	47.6	38.5
		MCWB	33.7	33.6	40.9	47.2	53.9	59.6	64.1	61.5	56.4	48.5	39.6	34.1
	10%	DB	33.3	33.6	44.5	59.0	70.4	75.4	81.7	82.1	73.3	57.1	42.2	34.1
		MCWB	30.6	31.0	38.0	44.5	51.8	58.0	62.7	60.5	54.3	45.6	36.5	31.1
	0.4%	WB	39.5	39.4	48.8	54.4	60.3	67.1	70.7	67.4	62.2	56.1	47.8	41.3
		MCDB	45.0	44.1	61.0	71.8	76.3	80.5	86.1	83.4	83.2	73.0	59.6	47.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.0	35.7	44.7	51.1	57.3	63.8	67.9	65.3	59.6	52.3	43.6	37.0
		MCDB	39.5	38.9	54.1	67.0	76.1	77.0	83.2	81.5	80.2	65.8	52.3	41.8
	5%	WB	33.8	33.7	41.3	48.2	55.2	61.5	65.9	63.6	57.4	49.2	40.0	34.2
		MCDB	36.9	36.7	48.8	62.3	72.1	74.8	80.7	79.3	75.2	60.6	46.6	38.4
	10%	WB	30.6	31.0	38.2	45.5	53.1	59.4	64.2	61.8	55.2	46.4	36.8	31.1
		MCDB	33.3	33.5	44.2	57.4	68.1	71.7	77.8	77.5	71.3	56.3	41.5	34.2
Mean Daily Temperature Range	5% DB	MDBR	19.0	19.3	19.9	24.5	25.5	23.0	26.1	27.3	26.3	23.6	19.2	18.4
		MCDBR	20.9	19.9	27.3	35.2	34.3	30.6	33.7	36.7	37.9	32.9	26.1	21.1
		MCWBR	17.4	16.1	16.7	18.5	15.4	13.0	13.4	13.7	16.4	18.1	17.0	16.2
	5% WB	MCDBR	21.2	19.6	25.8	32.0	29.7	26.0	28.7	29.6	33.5	30.8	24.5	20.8
		MCWBR	18.2	16.1	16.7	17.6	13.8	12.3	13.1	12.3	15.2	17.6	16.5	16.2
Clear-Sky Solar Irradiance	taub		0.271	0.262	0.292	0.351	0.376	0.369	0.376	0.393	0.338	0.298	0.267	0.246
	taud		2.321	2.437	2.452	2.297	2.271	2.355	2.357	2.265	2.406	2.495	2.470	2.412
	Ebn at Noon		242	283	292	282	279	281	277	265	271	265	247	239
	Edh at Noon		20	23	28	37	40	37	37	38	30	22	17	16
All-Sky Solar Radiation	RadAvg		377	701	1131	1535	1851	1942	2100	1697	1232	752	400	282
	RadStd		43	63	86	117	125	141	111	132	108	90	39	31

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	-0.68	N/A	N/A	N/A	N/A	N/A
(47)	Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page